

EECS 4422

SUMMER 2026 CLASS PROJECT

A Comparative Study of SIFT and ORB for Wide-Baseline Feature Matching

Github Link: <https://github.com/ThorOdinCode/Computer-Vision-Project>

THOR LASKI | EMILY ZELKOWICZ | ANTONIO LAMACCHIA |
EKAMBIR MOMI | ABRAR TARAFDER



Introduction

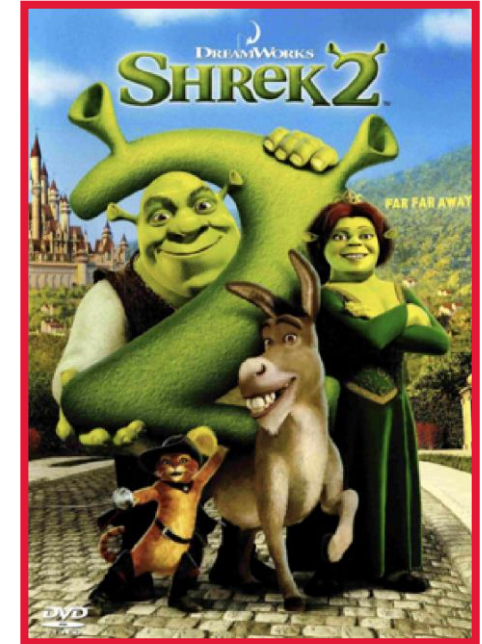
› Local Feature Matching

- Local feature matching is a fundamental problem in Computer Vision
 - Identifies corresponding points between images of the same scene
 - Enables many Computer Vision applications:
 - Image Registration - Aligning two images of the same scene.
 - Panorama Stitching - Combining multiple images into one wide image.
 - Visual Localization - Determining a camera's position in an environment.
 - Structure-from-Motion (SfM) - Reconstructing a 3D scene from multiple images.
 - Simultaneous Localization and Mapping (SLAM) - Building a map while tracking the camera's location.
 - **Wide Baseline Feature Matching is an extension of Local Feature Matching where images are captured from significantly different viewpoints.**
-
- ## › Our Focus:
- Compare SIFT and ORB for wide-baseline feature matching using the HPatches benchmark.



Challenges To Consider

- **Large viewpoint changes make corresponding features difficult to identify.**
 - Objects appear very different when viewed from different angles.
- **Scale and rotation variations change the appearance of local features.**
 - Features may appear larger, smaller, or rotated between images.
- **Illumination changes affect feature detection and descriptor matching.**
 - Differences in lighting can make the same features look different.
- **Incorrect matches (outliers) reduce matching accuracy and geometric consistency.**
 - False feature correspondences reduce matching accuracy.
- **Trade-off between accuracy and speed is critical for real-world applications.**
 - More accurate methods are often slower, while faster methods may be less reliable.



Motivation For This Project

- **SIFT** is known for its robustness but has **higher computational cost and the feature extraction is a more manual process.**
- **ORB** is significantly faster but **may sacrifice matching accuracy.**
- It is important to understand which method performs better under challenging wide-baseline conditions.
- HPatchs provides a standardized benchmark with viewpoint and illumination sequences for fair evaluation.
- Goal: Compare SIFT and ORB in terms of accuracy, robustness, and runtime.



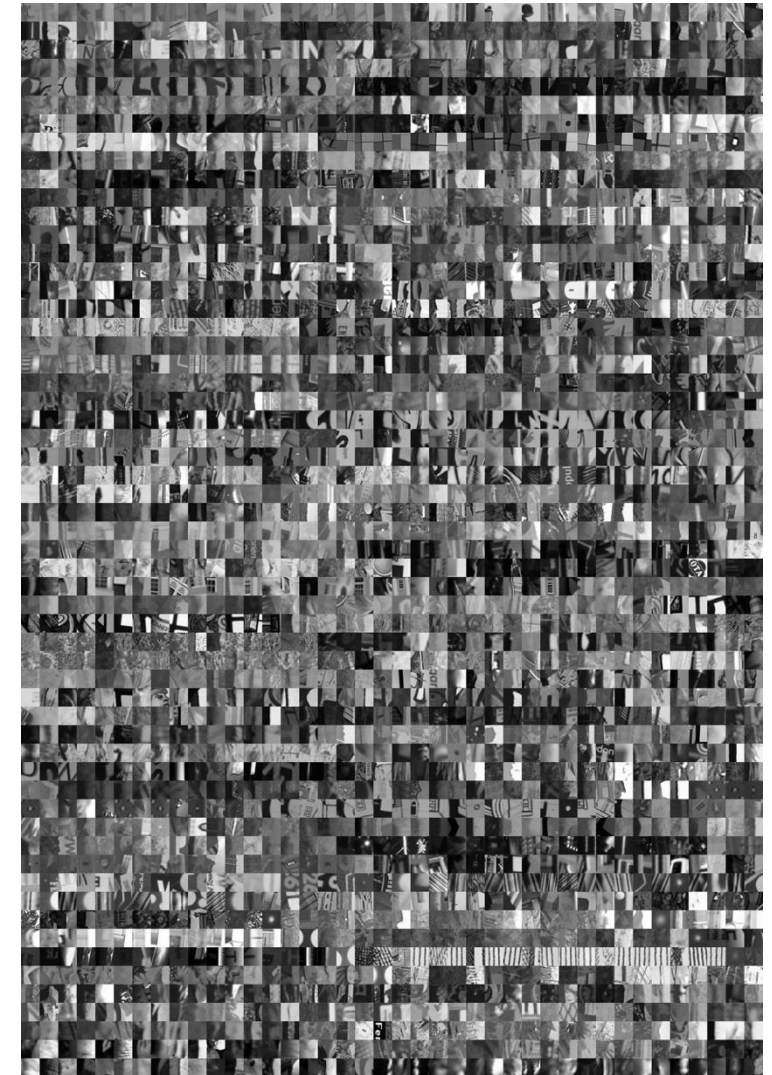
Dataset We Used

› This project uses the HPatches benchmark

- A widely used dataset for evaluating local feature detection and matching under challenging viewpoint and illumination changes.
- Dataset provides image sequences together with ground-truth homography matrices, enabling quantitative evaluation of local feature matching under viewpoint and illumination changes.

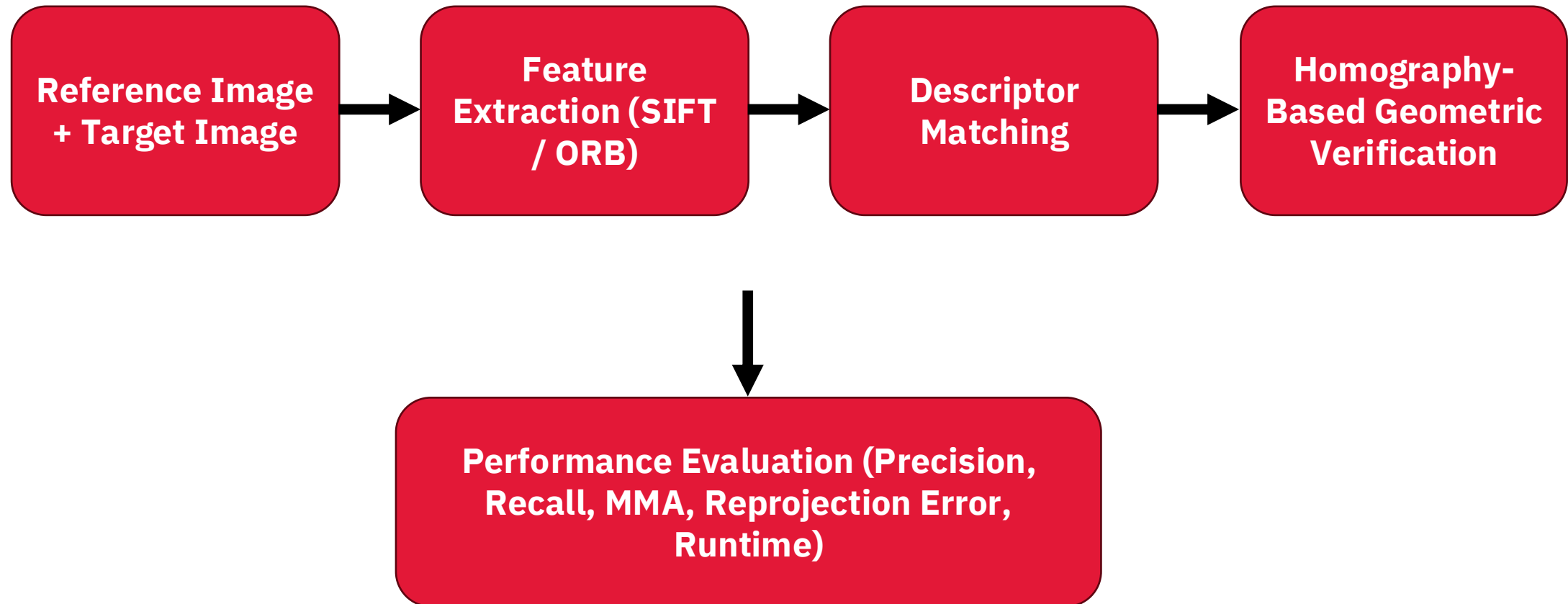
› HPatches Benchmark

- **HPatches benchmark consists of 116 image sequences**
 - Each sequence contains a reference image together with five transformed images generated from viewpoint or illumination changes.
- **The dataset contains two types of image sequences**
 - 59 v_sequences which introduce viewpoint changes
 - 57 i_sequences which introduce illumination changes
 - We used the HPatches-sequences-release dataset*
- › This combination of geometric and photometric variations makes HPatches a suitable benchmark for **evaluating the robustness** of local feature matching methods under challenging real-world conditions.



Methodology For Both Algorithms I

› Overall Pipeline



Methodology For Both Algorithms II

› Feature Extraction

• SIFT

- SIFT is designed to be robust to scale changes, in-plane rotation, and moderate illumination variation, which makes it well suited for feature-based matching across images captured under different conditions.
- **Step 1 Gaussian Scale Space:** Smooth the image at multiple scales: $L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$,
- **Step 2 Key point Detection:** Detect the Difference-of-Gaussians extrema: $D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$,
- **Step 3 Keypoint Refinement:** Remove low-contrast and edge responses.
- **Step 4 Orientation Assignment:** Compute gradient magnitude and orientation.
- **Step 5 Descriptor Generation:** Form a 128-dimensional descriptor from local gradient histograms.
- **Step 6 Descriptor Normalization:** Normalize the descriptor for illumination robustness.

Below is an example of SIFT being ran on 4 different images from the HPatches-sequence release under different illumination conditions. The image bordered with red borders is the reference image and the other (blue borders) are test images. For reference, a maximum of 150 points are shown for ease of visualization.



Methodology For Both Algorithms II

› Feature Extraction

• SIFT (continued)

- › At the implementation level, we used OpenCV's built-in SIFT detector.
- For each image, SIFT returns a set of keypoints and a 128-dimensional descriptor for every keypoint.
- In each HPatches sequence, image 1 is the reference image and images 2 through 6 are test images.
- Feature extraction is run on every image in the sequence, not only on one pair.
- No extra preprocessing is used beyond converting the image to grayscale.
- SIFT is strong under scale changes, rotation, and moderate illumination changes.
- SIFT is slower than ORB because the descriptors are larger and use floating-point values.
- SIFT can struggle in low-texture regions, repetitive patterns, or under extreme viewpoint change.



Methodology For Both Algorithms II

- **Feature Extraction**

- **ORB**

- ORB is designed to provide fast, rotation-aware feature extraction that uses efficient binary descriptors which makes it suitable for real-time feature matching.
 - **Step 1 Image Pyramid:** Construct multiple image levels to detect features at different scales.
 - **Step 2 FAST Keypoint Detection:** Identify candidate corners by comparing pixels around each candidate point.
 - **Step 3 Keypoint Selection:** Rank detected keypoints and retain the strongest responses.
 - **Step 4 Orientation Assignment:** Estimate each keypoint's orientation using the intensity-centroid method.
 - **Step 5 Descriptor Generation:** Apply an orientation-aware BRIEF test to generate a 256-bit binary descriptor.
 - **Step 6 Binary Descriptor Output:** Store each descriptor in 32 bytes to enable efficient comparison using Hamming distance.

Below is an example of ORB from the HPatches sequence under different illumination conditions. The blue is the reference image, while the remaining three are test images. The green circles are the scale and location of the strongest ORB keypoints, and the red dots are the centers. A maximum of 150 keypoints per image is shown.



Methodology For Both Algorithms II

- **Feature Extraction**

- **ORB (continued)**

- For the implementation, we use OpenCV's built-in ORB detector.
 - Each image is converted from RGB to grayscale before feature extraction.
 - OpenCV's `detectAndCompute()` function performs keypoint detection and descriptor generation together.
 - For each image, ORB returns a set of keypoints and a 256-bit binary descriptor for each keypoint.
 - ORB uses the same `FeatureDetector` interface as SIFT so both algorithms can receive the same image inputs.
 - Detected keypoints are visualized with red circles that show their position, scale, and orientation.
 - ORB is faster and more memory-efficient than SIFT because it uses compact binary descriptors.
 - ORB is considered less accurate than SIFT with viewpoint changes and struggles with low-texture regions.



Methodology For Both Algorithms III

› Feature Matching

• SIFT

- In the Hatches dataset, the **first image in each sequence is used as the reference image.**
- Each remaining image is treated as a test image.
- Matching is performed one test image at a time, so **only two images are compared in each pair.**
- SIFT descriptors are 128-dimensional feature vectors.
- Euclidean distance is used to find the nearest descriptor matches. $\Delta_{ij} = \|\mathbf{d}_i - \mathbf{q}_j\|_2$.
- Lowe's ratio test is applied to reject ambiguous or weak matches. $r_i = \frac{\Delta_{i,1}}{\Delta_{i,2}} < \tau$, where $\Delta_{i,1}$ and $\Delta_{i,2}$ are distances to the closest and second-closest descriptors and τ is set between 0.7 and 0.8
- mutual-best matching can be used to verify matches geometrically and remove outliers.
- The **final inlier matches provide reliable correspondences** for image alignment and other downstream tasks.

An example of SIFT Feature Matching between the first image (reference image) and another image from the set of images (a test image).

For sampling purposes, only top ten matches are shown →



Methodology For Both Algorithms III

➤ Feature Matching

• SIFT

- Matching uses OpenCV's brute force matcher with $k=2$ for nearest neighbours
- For each reference descriptor, we look at the best and second-best match, plugging each into Lowe's ratio test
- Mutual-best consistency checks describe a reverse matching step from the test image back to the reference image
- Matches passing the ratio test and mutual-best check improve robustness, discarding weak correspondences
- This stage does not warp the image or estimate an affine / homography model
- These correspondences are later checked with the ground-truth homography, and only the geometrically correct ones are counted



Methodology For Both Algorithms III

› Feature Matching

- ORB
 - › In each HPatches sequence, the first image is used as the reference image.
 - The remaining images are treated as a test image and compared with the reference separately.
 - ORB creates 256-bit binary descriptors that are stored by OpenCV as 32-byte vectors.
 - Descriptor similarity is measured using Hamming distance, which counts the number of different bits:
 - A lower Hamming distance means a more similar pair of descriptors.
 - Lowe's ratio test rejects unclear matches by comparing the closest and second-closest candidates
 - Mutual-best matching keeps only correspondences that agree in both matching directions.
 - The remaining matches are later checked using the ground-truth homography to identify geometrically correct correspondences.
 - An example of ORB Feature Matching between the first image (reference image) and another image from the set of images (a test image).

For sampling purposes, only top ten matches are shown



Methodology For Both Algorithms III

➤ Feature Matching

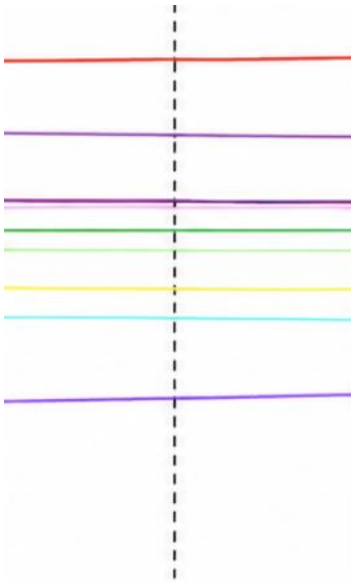
• ORB (continued)

- Images are converted to grayscale before ORB detection and descriptor begins the computation
- OpenCV's ORB detector is configured to extract 100 features per image or less
- The implementation uses a scale factor of 1.2, 8 pyramid levels, Harris scoring, and a FAST threshold of 20.
- Matching should use OpenCV's brute-force matcher with NORM_HAMMING and k=2 nearest neighbours.
- Every reference descriptor has its closest and second-closest test descriptors that are passed through Lowe's ratio test.
- A mutual-best check can then retain only matches that are also the best correspondence in the reverse direction.
- This stage only establishes descriptor correspondences but doesn't change the images or estimate a new homography.
- The correspondences that are accepted are evaluated against the HPatchs ground-truth homography, where reprojection errors determine which matches are correct.
- Matches are sorted by increasing Hamming distance, and only the strongest matches are displayed.



ORB vs SIFT Direct Matching Comparison Example

SIFT:



The SIFT results identify a larger number of reliable correspondences distributed across the facade, including windows, columns, and roof features. The ORB results produce accurate correspondences on several prominent corners of the building but are **limited in coverage due to the lower distinctiveness of binary descriptors.**

The distribution visualization on the right of each image highlights the spatial coverage of matched features across the image pair. SIFT produces a broader and denser distribution of correspondences over distinctive architectural structures, while ORB yields a sparser distribution concentrated on strong corners and edges.

ORB:



SIFT demonstrates greater robustness in repetitive architectural scenes because its gradient-based descriptors better distinguish similar-looking structures, resulting in more reliable feature matching overall.

Methodology For Both Algorithms III

› Best Correspondence Selection between SIFT and ORB Methodology

- Correspondence refers to a **matched pair of keypoints**, one in the reference image and one in the target image, that are estimated to represent the same physical scene point (barrier b/w reference and target in red). The below images are an example of corresponding keypoints that are matched through SIFT. The same can apply to ORB.



- **Filter ambiguous matches:** Usually correspondence selection applies Lowe's ratio test followed by a mutual-best (cross-check) consistency check to retain only reliable correspondences.
- **Select best matches:** After that, the remaining matches are ranked by descriptor distance and keep the strongest correspondences for visualization and geometric verification.

Between ORB and SIFT

- **SIFT has the best correspondence selection overall in this project.**
 - It produces more distinctive matches, so fewer incorrect correspondences survive the ratio test and geometric filtering. ORB is faster and lighter, but its binary descriptors are usually less discriminative, so match quality is typically lower.
- This means that **SIFT is used when accuracy matters, and ORB is used when speed matters.**

Methodology For Both Algorithms IV

› Homography-Based Geometric Verification

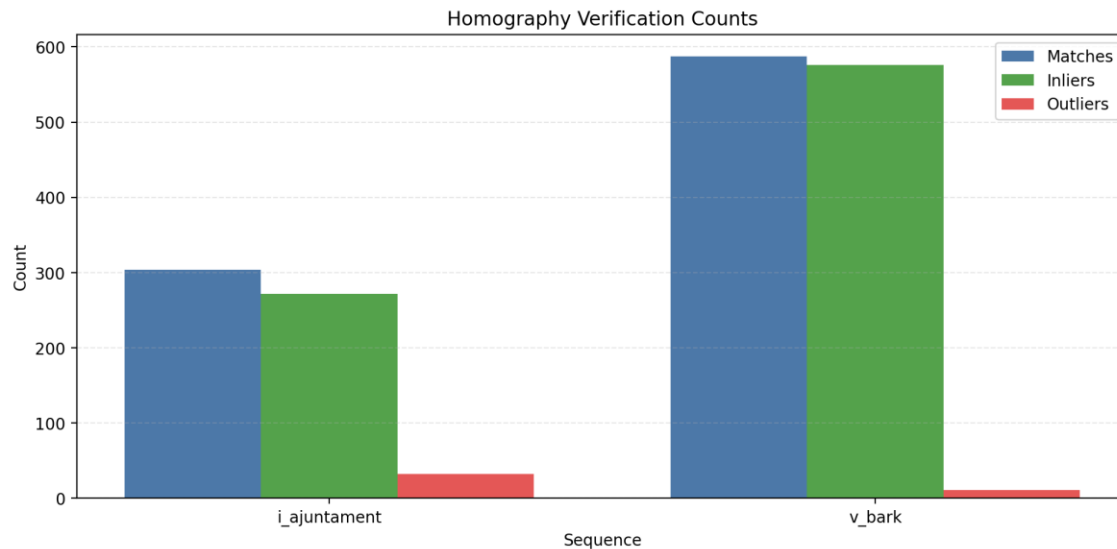
- The ground-truth HPatches homography maps each matched keypoint from the reference image into the test image.
- For a reference keypoint $x = [x, y, 1]^T$, the expected position is $\hat{x}' \sim Hx$
- Homogenous coordinates are normalized to get the predicted image coordinates: $\hat{x}' = u/w, \hat{y}' = v/w$
- The reprojection error is the Euclidean distance between the predicted location and the location of the matched test keypoint: $e_i = ||\hat{x}'_i - x'_i||_2$
- A correspondence is classified as a geometrically correct match when its reprojection error is at most 3 pixels.
- Matches with errors greater than 3 pixels are considered outliers.
- The ground-truth homography is used only to verify correspondences and the code does not estimate a new homography or warp either image.
- The resulting inlier counts and reprojection errors are used to calculate precision, recall, and Mean Matching Accuracy (MMA).
- MMA is evaluated at thresholds of 1, 2, 3, 5, 8, and 10 pixels which allows matching accuracy to be seen under different geometric tolerances.

Methodology For Both Algorithms IV

› Homography-Based Geometric Verification: Results

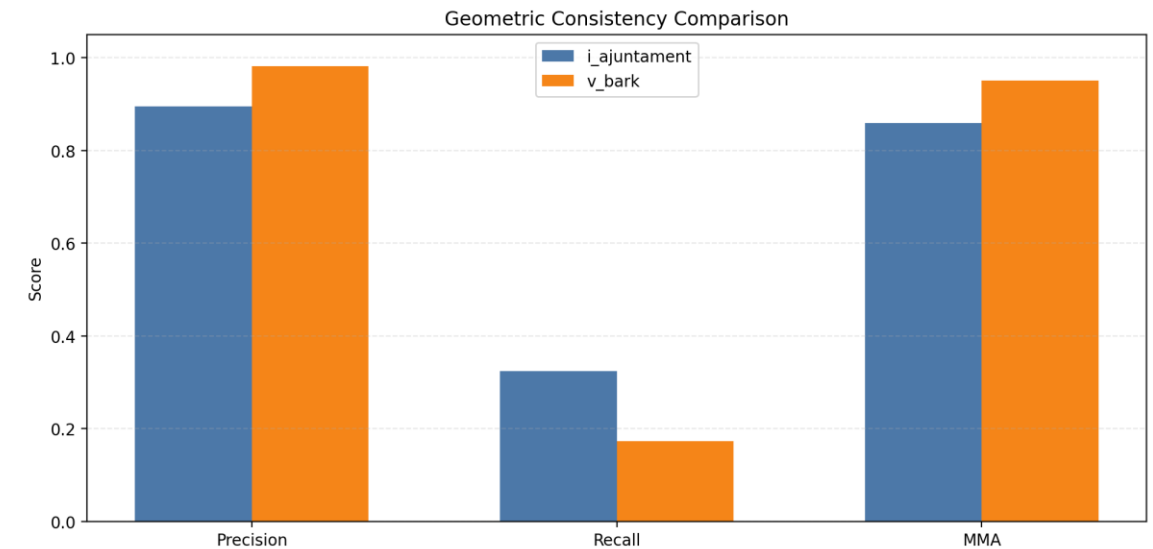
• Match Counts:

- i_ajuntament: ~305 matches (~272 inliers, 33 outliers)
- v_bark: ~588 matches (~577 inliers, only 11 outliers)
- Viewpoint matching: more matches AND cleaner



• Precision, Recall, MMA:

- Precision: v_bark ≈ 0.98 vs. i_ajuntament ≈ 0.89
- MMA: v_bark ≈ 0.95 vs. i_ajuntament ≈ 0.86
- Recall flips: i_ajuntament ≈ 0.32 vs. v_bark ≈ 0.17
- Why: viewpoint change pushes more keypoints out of frame $\rightarrow$ smaller "matchable" denominator

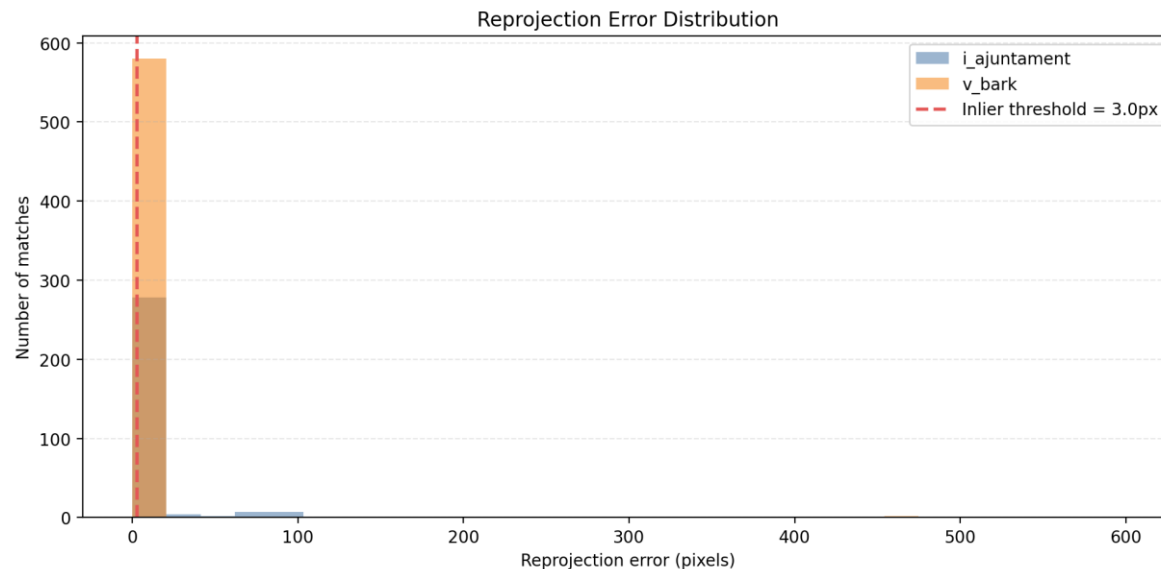


Methodology For Both Algorithms IV

› Homography-Based Geometric Verification: Results

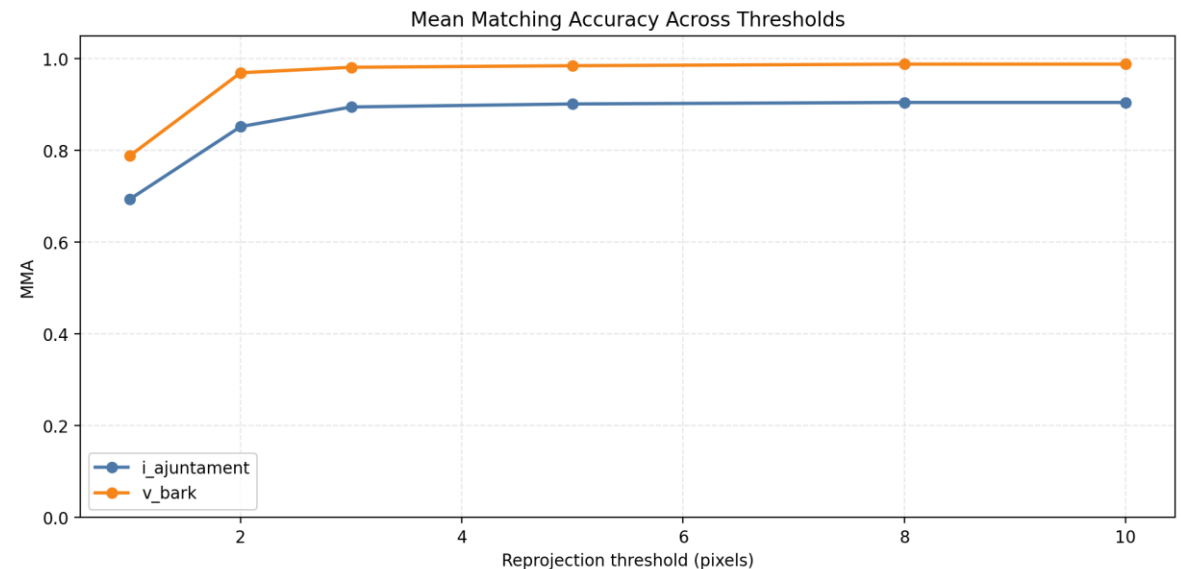
• Reprojection Error Detection:

- Both sequences cluster tightly near zero error, under 3px threshold
- *i_ajuntament*: small tail of outliers (~60–100 px) → lowers precision
- *v_bark*: tighter overall, almost all matches near zero



• MMA Curve:

- *v_bark* above *i_ajuntament* at every threshold
- Largest gap at 1px (≈ 0.79 vs. ≈ 0.69)
- Both plateau by ~3px → remaining gap = genuine outliers, not borderline matches



Methodology For Both Algorithms IV

› Homography-Based Geometric Verification: Interpretation

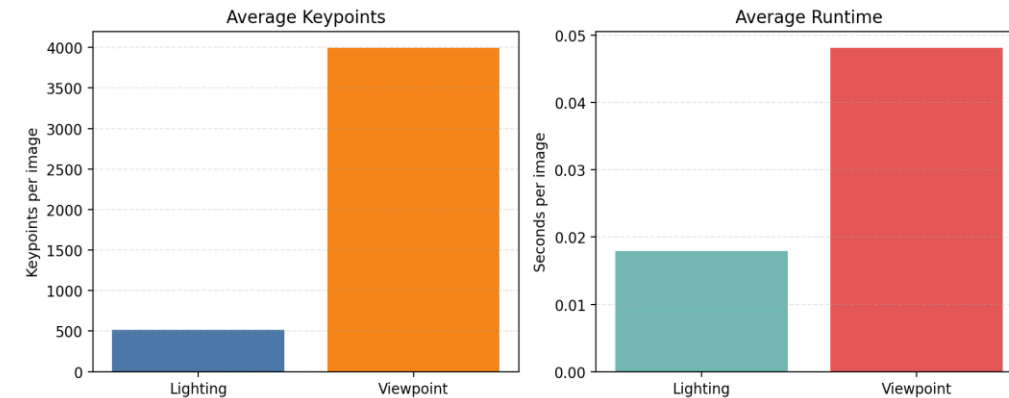
- Overall the viewpoint change were handled *more* precisely/accurately than illumination change — counterintuitive for SIFT
- Likely cause: v_bark's scene texture → strong, stable keypoints under perspective shift
- i_ajuntament's lighting variation may destabilize some descriptors → longer error tail
- Precision/recall trade-off is invisible in raw match counts — this is why these metrics matter for the upcoming SIFT vs. ORB comparison

Evaluation I

➤ Feature Detection Comparison For SIFT

• Number of Detected Keypoints

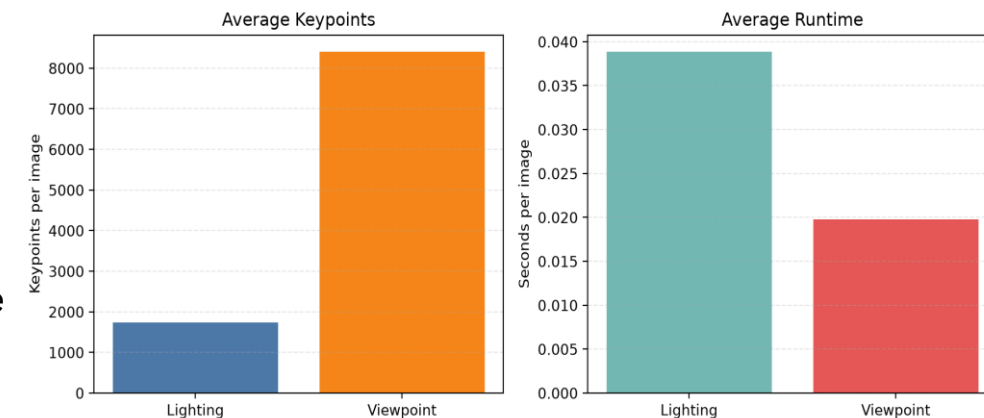
- SIFT keypoint density varies significantly with image texture and lighting conditions.
- Highest-density illumination sequences: *i_dc*, *i_kurhaus*, and *i_indiana*.
- Highest-density viewpoint sequences: *v_busstop*, *v_sunseason*, and *v_london*.
- Severe illumination changes reduce the number of detected keypoints.
- Highly textured viewpoint scenes retain more keypoints but require longer extraction times.



➤ Feature Detection Comparison For ORB

• Number of Detected Keypoints

- Measures how many local features ORB detects in each HPatches image.
- ORB is configured to have a maximum of 10000 keypoints per image.
- The keypoints are ranked using the Harris corner response, and only the strongest responses are considered.
- Results are grouped into illumination-change and viewpoint-change sequences.
- A constant and high keypoint count means that ORB continues to detect distinctive image structures as conditions change.
- A decrease in keypoints suggests that illumination or viewpoint changes make corners difficult to detect.
- Keypoint count measures feature-detection coverage, yet does not indicate whether the detected features can be matched correctly.
- These results should be interpreted alongside runtime, precision, recall, and MMA.



Evaluation II

› Feature Matching Results for SIFT

• Precision

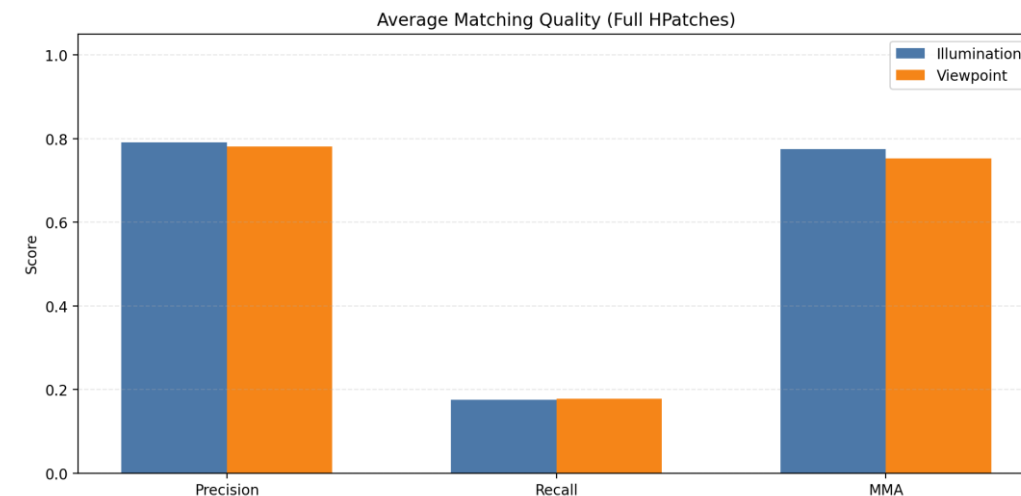
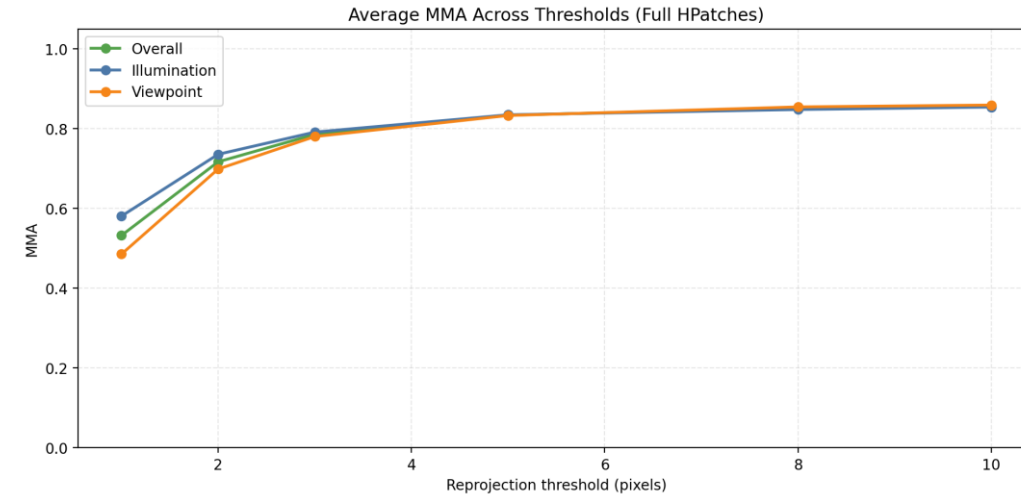
- Measures the proportion of accepted SIFT matches that are geometrically correct.
- Average precision: 0.786 (Illumination: 0.791, Viewpoint: 0.780).
- High on easy pairs (~ 0.90 for $1 \rightarrow 2$) and decreases on difficult pairs (~ 0.69 for $1 \rightarrow 6$).
- Most retained matches are within 3 pixels of the ground-truth homography.

• Recall

- Measures the proportion of visible reference keypoints that are correctly matched.
- Average recall: 0.177 (≈ 0.29 for $1 \rightarrow 2$, ≈ 0.10 for $1 \rightarrow 6$).
- Strict ratio and mutual-best filtering remove many candidate matches.
- Indicates high match quality but limited feature coverage.

• Mean Matching Accuracy (MMA)

- Measures the fraction of matches below a given reprojection error threshold.
- Average MMA: 0.763 (Illumination: 0.774, Viewpoint: 0.752).
- Increases from ~ 0.53 (1 px) to ~ 0.79 (3 px) and plateaus at ~ 0.85 – 0.86 (8–10 px).
- Declines with increasing difficulty (0.875 for $1 \rightarrow 2 \rightarrow 0.674$ for $1 \rightarrow 6$), showing reduced localization accuracy.



Evaluation II cont'd

› Feature Matching Results for ORB

• Precision

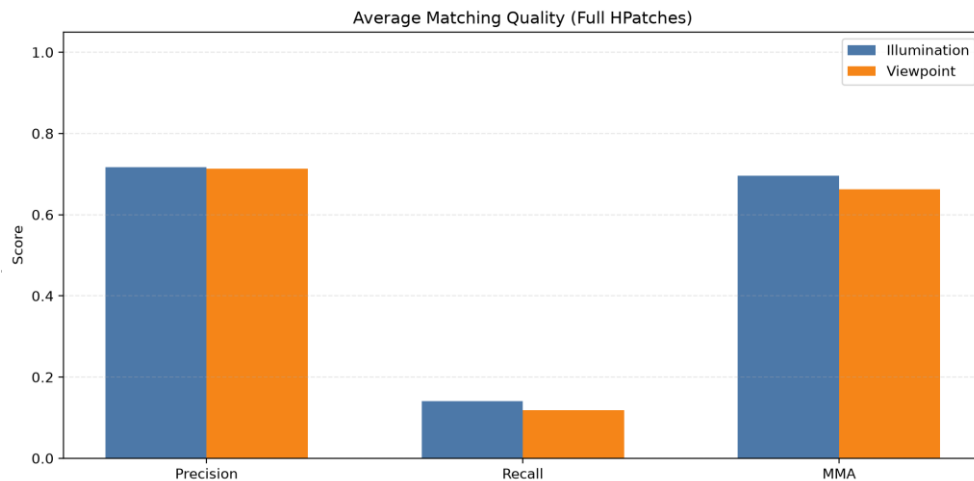
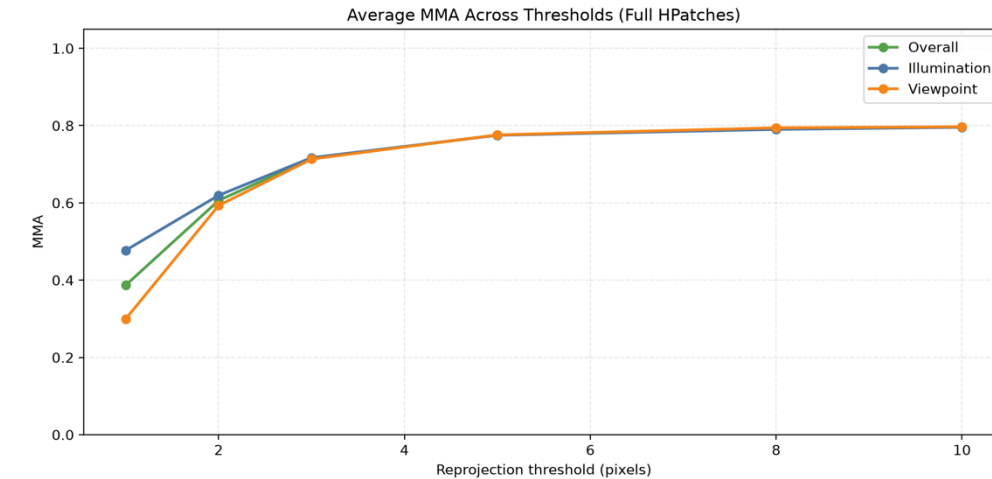
- Measures the proportion of filtered ORB matches that are geometrically correct within 3 pixels.
- Average precision: 0.754 (Illumination: 0.777, Viewpoint: 0.731).
- High on easy pairs (~0.898 for 1→2) and decreases on difficult pairs (~0.625 for 1→6).
- Most retained matches are within 3 pixels of the ground-truth homography.

• Recall

- Measures the proportion of visible reference keypoints recovered as correct matches.
- Average recall: 0.124 (≈0.223 for 1→2, ≈0.066 for 1→6).
- Ratio-test and mutual-best filtering remove many ambiguous candidates
- High precision but low recall indicates reliable matches with limited feature coverage.

• Mean Matching Accuracy (MMA)

- Measures the fraction of matches below a range of reprojection-error thresholds
- Average MMA: 0.717 (Illumination: 0.75, Viewpoint: 0.684).
- Increases from ~0.51 (1 px) to ~0.777 (3 px) and levels off at 5px.
- Declines with increasing difficulty (0.848 for 1→2 → 0.601 for 1→6), showing reduced localization accuracy.
- This indicates that most retained ORB matches were geometrically accurate



Evaluation III

› Robustness Results for SIFT

• Reprojection Error

- Measures the geometric distance between a matched test-image keypoint and the position predicted by the HPatchs ground-truth homography.
- Most SIFT matches have low reprojection error, indicating accurate geometric alignment.
- Most correct matches fall within 3 pixels of the ground-truth homography.
- Higher mean error is mainly caused by a small number of outlier matches.

• Viewpoint Changes

- Evaluates SIFT under changes in camera angle, scale, perspective, and visible image content.
- Viewpoint sequences produce more matches but slightly lower MMA (0.752) than illumination sequences (0.774).
- Larger viewpoint changes mainly reduce recall and the number of valid matches.
- Strong performance on textured scenes (e.g., v_bark) is not representative of the overall dataset.

• Illumination Changes

- Evaluates SIFT with brightness, contrast, shadows, or exposure changes while the viewpoint is approximately fixed.
- Severe lighting changes reduce keypoint detections and total matches.
- Remaining matches are still highly accurate despite fewer detections.
- Overall, illumination sequences achieve slightly higher MMA than viewpoint sequences.

Evaluation III Cont'd

› Robustness Results for ORB

• Reprojection Error

- Measures the distance between a matched keypoint and its position predicted by the HMatches ground-truth homography.
- Matches with an error of 3 pixels or less are classified as inliers; matches above 3 pixels are outliers.
- Lower reprojection error indicates greater geometric consistency.
- Most ORB matches are geometrically correct within 3 pixels, with MMA largely plateauing after about 5 pixels.

• Viewpoint Changes

- Across the full dataset, viewpoint sequences achieved 0.731 precision, 0.109 recall, and 0.684 MMA.
- Performance decreases further under extreme viewpoint changes, particularly for harder 1→5 and 1→6 pairs.
- The v_bark 1→2 example achieved high precision but low recall, showing reliable matches but limited feature recovery.
- v_bark is a favorable textured example and is not representative of the overall viewpoint results.

• Illumination Changes

- Across the full dataset, illumination sequences achieved 0.777 precision, 0.140 recall, and 0.750 MMA, outperforming viewpoint sequences.
- ORB remains relatively robust to brightness, contrast, shadow, and exposure changes when viewpoint is approximately fixed.
- Strong illumination changes can still alter FAST corner responses and BRIEF intensity comparisons, reducing matching accuracy.

Evaluation IV

› Performance Analysis for SIFT vs. ORB

• SIFT

- **SIFT has a higher computational cost** because each detected keypoint is represented using a 128-dimensional floating-point descriptor.
- Feature matching uses Euclidean distance between descriptors, requiring **more expensive floating-point calculations** compared with ORB's binary matching.
- SIFT runtime increases with keypoint density, with the full feature extraction and matching pipeline taking approximately 1 second per image pair in our evaluation.
- The additional computation and descriptor memory make SIFT less suitable for real-time applications, but support its stronger matching robustness.

• ORB

- ORB achieves **greater computational efficiency** by combining FAST keypoint detection with compact 256-bit binary descriptors.
- ORB descriptors require only 32 bytes per keypoint, reducing descriptor storage and memory requirements compared with SIFT's floating-point representation.
- Feature matching uses Hamming distance, which compares binary descriptors using efficient bitwise operations rather than floating-point Euclidean-distance calculations.
- The lower computational and memory requirements make **ORB better suited for real-time and resource-constrained applications.**

Evaluation IV

➤ Final Discussion

- **SIFT prioritizes matching accuracy over speed**, achieving higher overall precision (0.786), recall (0.177), and MMA (0.763) than ORB (0.754 precision, 0.124 recall, and 0.717 MMA).
- **SIFT performance decreases as image difficulty increases**, with MMA dropping from 0.875 for easier 1→2 pairs to 0.674 for harder 1→6 pairs, while illumination and viewpoint sequences show similar overall performance.
- **SIFT is computationally expensive** because each keypoint is represented by a 128-dimensional floating-point descriptor, requiring more memory and costly Euclidean-distance computations during feature matching.
- **ORB is faster and more memory-efficient** due to FAST keypoint detection and compact 256-bit binary descriptors, reducing the computational and storage requirements of feature extraction and matching.
- ORB uses Hamming distance, enabling efficient bitwise operations instead of floating-point computations during matching, but its **matching performance decreases more strongly on difficult image pairs**, with MMA dropping from 0.848 to 0.601 between 1→2 and 1→6.
- **ORB is well suited for real-time and resource-constrained applications** but is **less robust than SIFT to large viewpoint and appearance changes**, particularly because its image-pyramid approach provides more limited scale and viewpoint invariance.
- Overall, **SIFT is preferable when matching robustness and geometric accuracy are the priority**, while **ORB is preferable when speed, memory efficiency, and real-time performance are more important**.

References and Citations

- Vassileios Balntas, Karel Lenc, Andrea Vedaldi, and Krystian Mikolajczyk. Hpatches: A benchmark and evaluation of handcrafted and learned local descriptors. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pages 5173–5182, 2017.
- David G. Lowe. Distinctive image features from scale-invariant keypoints. International Journal of Computer Vision, 60(2):91–110, 2004.
- Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. Orb: An efficient alternative to sift or surf. In Proceedings of the IEEE International Conference on Computer Vision, pages 2564–2571, 2011.